SYSTEMS, EACH WITH ITS CHOSEN SOLUTION AND SETTINGS

foundation: posts-at-corners (solution posts-at-corners)
foundation.embedment_mm = 1219 mm; no reason given
foundation.above_grade_mm = not given mm; reason: the sketch's '?' from grade to the floor; Jim's word gives it
foundation.section_mm = not given mm; reason: the sketch draws no post size; Jim's choice or a load check gives it
foundation.frost_line_mm = 1067 mm; reason: 42 in, Michigan frost depth, research row site.frost_mrc
floor: platform (solution platform)
floor.width_mm = 2438 mm; no reason given
floor.depth_mm = 2438 mm; no reason given
floor.thickness_mm = 114 mm; reason: assumed: a 4-1/2 in (114 mm) SIP floor panel, the wall standard's thickness, research row sip.prs610_panel_thickness; Jim's floor build-up gives it
floor.elevation_mm = -114 mm; reason: the floor's top is the datum the walls and their 12 ft stand on, so its 114 mm panel lies below it
walls: walls-on-edges (solution walls-on-edges)
walls.height_mm = 3658 mm; reason: reading: the south elevation's '12' as wall height, floor top to roof underside; 12 ft overall above grade is the other reading
walls.thickness_mm = 114 mm; reason: a 4-1/2 in (114 mm) SIP wall panel, ANSI/APA PRS 610.1 sec. 4.1, research row sip.prs610_panel_thickness
walls.panel_width_mm = 1219 mm; reason: 4 ft is a sold width (sip.stock.premier_widths, sip.stock.acme_sizes); chosen narrow for handling and take-down; open for ranking
roof: flat-roof-over (solution flat-roof-over)
roof.width_mm = 3658 mm; no reason given
roof.depth_mm = 5486 mm; no reason given
roof.west_eave_mm = 610 mm; no reason given
roof.north_eave_mm = 610 mm; reason: reading: the hatched hangover on the south, 8 ft past the floor; centred, 5 ft each end, is the other reading
roof.thickness_mm = not given mm; reason: the sketch draws no rafter or deck; Jim's framing gives it
glazing: Two 4 ft by 8 ft openings stacked over the knee wall (solution two-stacked-4x8)
glazing.sides = south compass_sides; reason: Jim, 2026-10-18: only south-facing glass; no east, west or north window
glazing.first_sill_mm = 1219 mm; reason: Jim: the first 4 ft of the south wall is a SIP panel below the windows
glazing.glazed_height_mm = 2438 mm; reason: Jim: two stacked 4x8 windows, the upper one lighting the sleeping loft, so 8 ft of glass over the knee wall; the other window-wall options (one-8x8, four-4x4) fill the same 8 ft
weather: conditions (solution conditions)
weather.u_value = 2.8 w_per_m2_k; reason: assumed: typical double glazing, not chosen by Jim
weather.solar_heat_gain_coefficient = 0.6 ratio; reason: assumed: typical double glazing, not chosen by Jim
weather.inside_celsius = 20 celsius; reason: assumed design day
weather.outside_celsius = 0 celsius; reason: assumed design day
weather.sun_watt_per_square_metre = 500 w_per_m2; reason: assumed clear-sky peak; the solver applies it to every side
finish: uniform (solution uniform)
finish.mean_reflectance = 0.5 ratio; reason: assumed mid-grey interior; Jim's wall finish gives it
finish.glass_transmittance = 0.68 ratio; reason: the BRE standard figure for double glazing as the Camden daylight reports quote it
finish.sky_angle_degrees = 65 degree; reason: assumed open sky; the site gives it
envelope: insulated-box (solution insulated-box)
envelope.wall_r = 20 r_value; reason: assumed R-20 for a SIP wall (ft2 F h per BTU); the SIP thickness gives it
envelope.roof_r = 30 r_value; reason: assumed R-30; the roof build-up gives it
envelope.floor_r = 20 r_value; reason: assumed R-20; the floor build-up gives it
envelope.door_r = 5 r_value; reason: assumed R-5 for an insulated door, not checked against a source
climate: site-normals (solution site-normals)
climate.heating_degree_days_f = 7104.1 degree_day_f; reason: NOAA NCEI 1991-2020 annual normal, base 65F, station USW00014850 TRAVERSE CITY CHERRY CPTL AP MI, read from the NCEI data service 2026-10-17
climate.north_glass_sun_btu_per_ft2_season = 32480 btu_per_ft2; reason: PVGIS-NSRDB hourly irradiance on a vertical plane at 44.7408 N, 85.5825 W, 2005-2015 average, October to April summed (aspect 180 = north, which PVGIS echoed as -179: 102.5 kWh/m2 = 32480 BTU/ft2), gross: no shading, counts sun on days that need no heat
climate.east_glass_sun_btu_per_ft2_season = 89761 btu_per_ft2; reason: PVGIS-NSRDB hourly irradiance on a vertical plane at 44.7408 N, 85.5825 W, 2005-2015 average, October to April summed (aspect -90 = east: 283.2 kWh/m2 = 89761 BTU/ft2), gross: no shading, counts sun on days that need no heat
climate.south_glass_sun_btu_per_ft2_season = 160451 btu_per_ft2; reason: PVGIS-NSRDB hourly irradiance on a vertical plane at 44.7408 N, 85.5825 W, 2005-2015 average, October to April summed (aspect 0 = south: 506.2 kWh/m2 = 160451 BTU/ft2), gross: no shading, counts sun on days that need no heat
climate.west_glass_sun_btu_per_ft2_season = 95438 btu_per_ft2; reason: PVGIS-NSRDB hourly irradiance on a vertical plane at 44.7408 N, 85.5825 W, 2005-2015 average, October to April summed (aspect 90 = west: 301.1 kWh/m2 = 95438 BTU/ft2), gross: no shading, counts sun on days that need no heat
heating: wood-boiler-hydronic (solution wood-boiler-hydronic)
heating.fuel_btu_per_unit = 20000000 btu; reason: Jim: one grown tree is about 20 million BTU, a cord of oak is 24 to 26 million by hobby sources
heating.boiler_efficiency = 0.7 ratio; reason: assumed 70 percent; sources run 55 to 80 percent
doors: door-on-side (solution door-on-side)
doors.width_mm = 1219 mm; reason: Jim, 2026-10-18: the north door's framing is 4 ft wide by 8 ft tall
doors.height_mm = 2438 mm; reason: Jim: 8 ft tall door framing; a 4 ft tall 8 ft wide SIP panel sits over the door and its neighbour
doors.sides = north compass_sides; no reason given
doors.offset_mm = 0 mm; reason: open: which end of the north wall the door is at; the west end is a placeholder
loft_floor: platform (solution platform)
loft_floor.width_mm = 2438 mm; no reason given
loft_floor.depth_mm = 2438 mm; no reason given
loft_floor.thickness_mm = not given mm; reason: the subfloor thickness is not banked; the deck thickness waits for the OSB choice
loft_floor.elevation_mm = 2438 mm; reason: Jim: the sleeping loft floor is 8 ft above the living floor
joists: joist-framing (solution joist-framing)
joists.member_width_mm = 38 mm; reason: a 2x6 is 1.5 by 5.5 in actual, 38 by 140 mm
joists.member_depth_mm = 140 mm; reason: assumed 2x6; the beam solver says whether it is enough
joists.spacing_mm = 406 mm; reason: 16 in on centre, assumed
joists.modulus_mpa = 9653 mpa; reason: E = 1.4 million psi for No.2 SPF, NDS 2018 Supplement Table 4A, research row lum.nds.fv_spf_no2 (visual-read, not binding); 1.4e6 psi x 0.006895 = 9653 MPa
joists.elevation_mm = 2438 mm; reason: under the loft deck
loads: occupancy-load (solution occupancy-load)
loads.live_load_psf = 30 psf; reason: 30 psf for sleeping rooms, research row site.live_sleeping
loads.dead_load_psf = 10 psf; reason: 10 psf, research row site.dead_ten_psf
egress_code: mrc-2015-egress (solution mrc-2015-egress)
egress_code.min_door_width_mm = 914 mm; reason: 3 ft, Michigan Residential Code 2015 R311.2, research row code.r311_2_door
egress_code.min_door_height_mm = 2032 mm; reason: 6 ft 8 in, same row
house: The house as the parts that bear on each other (solution load-bearing-parts)
panel: 4-1/2 in SIP to ANSI/APA PRS 610.1: EPS core between two 7/16 OSB facings (solution prs-610-sip-4-1-2)
lumber: Spruce-Pine-Fir No. 2 dimension lumber, NDS 2018 Table 4A (solution spf-no2)
header: Header of 1-3/4 by 11-7/8 in LVL plies, Microllam 2.0E-2600Fb (solution header-lvl-11-7-8)
header.plies = 2 count; reason: assumed: two plies side by side, 3-1/2 in of LVL or 3 in of 2x lumber, fit inside the SIP's 3-5/8 in core (sip.prs610_core_thickness)
roof_loads: The ground snow and the roof's dead load, on the walls the roof bears on (solution ground-snow-on-roof)
roof_loads.ground_snow_psf = 60 psf; reason: 60 psf, Grand Traverse County in the Michigan ground snow load table, research row gl.hdr.snow_grand_traverse; carried on the roof unreduced, as no banked row gives a flat-roof factor
roof_loads.roof_dead_load_psf = 15 psf; reason: assumed: 15 psf, the roof dead load Boise's roof-only header look-up assumes, research row gl.hdr.vl_lookup_roof_only; the roof build-up gives it
roof_loads.deflection_limit_ratio = 240 ratio; reason: assumed: span over 240, the live-load deflection limit of Boise's header tables, research row gl.hdr.vl_limits, applied to the whole load
roof_loads.roof_bears_on = south, north compass_sides; reason: assumed: the roof's 8 ft south hangover can only be carried by members running north-south across the south wall, so the roof bears on the south and north walls
labour: Hand-dug post holes and SIP panel setting at published rates (solution published-rates)
prices: ACME Panel's listed SIP blanks, 7/16 in OSB both sides on EPS (solution listed-sip-blanks)

SOLVERS, EACH ANSWER WITH WHAT IT RESTS ON

average-daylight-factor (light): known
daylight_factor_percent = 7.37
glazing-balance (thermal): known
heat_loss_watt = 332.86
solar_gain_watt = 1783.15
design-day-load (thermal): known
loss_btu_per_hour = 2076.43
solar_gain_btu_per_hour = 6084.37
heating-season (thermal): known
season_loss_btu = 9834109.4
season_solar_btu = 6159297.14
solar_share_percent = 62.63
fuel_units_needed = 0.26
joist-deflection (structural): known
deflection_mm = 4.26
span_over_deflection = 571.7
bending_stress_mpa = 4.65
gravity-load-path (structural): unknown
the ground reaction is unknown: post@3048,2438,-1219: not measured (size-unknown): no figure for width_mm, depth_mm, above_grade_mm, section_mm; 0 of 25 pieces carry a known force
pieces = 25
pieces_weight_known = 9
pieces_force_known = 0
sources: sip.prs610_panel_thickness (published); sip.prs610_core_thickness (published); osb.gp_7_16 (published); sip.sipa_definition (published); wt.syn_osb_sheet (inferred); e1295.core (published); lum.nds.fv_spf_no2 (visual-read); lum.nds.cf_2x6 (visual-read); lum.nds.cr (published)
header-check (structural): known
reference design values with no NDS adjustment factor; the roof carries the ground snow load unreduced; a header's load is the roof over its wall and the joists landing at its head, not its own weight nor the wall or glass above it
headers = 3
headers_over_one = 0
utilisation_max = 0.61
sources: lvl.width (published); lvl.depth (published); gl.prop.lvl_2_0e_2600 (visual-read); gl.prop.lvl_density_unpublished (unpublished)
work-at-published-rates (tasks): known
post_holes_hours = 2
floor_hours = 1
walls_hours = 5.72
roof_hours = 1.91
measured_hours = 10.63
panel_usd = 3930.23
pieces_not_priced = 5
not_measured = set-posts (task.t2.hours_set_post_unpublished); lay-subfloor (task.t4.hours_subfloor_unpublished); cut-openings (task.t6.hours_field_opening_unpublished); set-windows-and-door (task.t7.hours_windows_door_unpublished); frame-loft (task.t9.hours_loft_measured_unpublished); install-boiler (task.t11.hours_boiler_hydronic_total_manhours_unpublished)
sources: task.t1.hours_hand_ramirent (published); task.t5.hours_premier_cs1_sqft (inferred); task.t5.hours_premier_cs1_sqft (published); task.t8.hours_premier_cs3_roof_rate (published); sp.acme_other_thicknesses (published); sp.acme_1025_price (published)

RULES, EACH WITH ITS STATUS

posts-under-floor [foundation]: PASS: every floor corner has a post under it; source: integrity: a floor on posts needs a post at each corner
walls-enclose-floor [walls]: PASS: walls close every edge of the floor; source: integrity: a shell closes every edge of its floor
roof-is-over-floor [roof]: PASS: a roof stands over the whole floor; source: integrity: a roof stands over the floor it keeps dry
egress-door-size [doors]: PASS: the egress door is at least as large as the limit; source: Michigan Residential Code 2015 R311.2, research row code.r311_2_door: side-hinged, 3 ft by 6 ft 8 in
posts-below-frost-line [foundation]: PASS: the posts are buried below the frost line; source: assumed: the frost depth the site gives; a footing must sit below it
roof-covers-floor [roof]: PASS: the roof covers the floor; source: Jim's what: the roof keeps the floor dry
headers-within-capacity [header_check]: PASS: every header is within its capacity; source: the header's catalog design values: bending, shear and its deflection limit
joists-span-the-deck [joists]: PASS: the joists span the deck at the stated spacing; source: integrity: a deck needs joists from edge to edge at the stated spacing

GRAVITY LOAD PATH, PER POST

status: unknown; the ground reaction is unknown: post@3048,2438,-1219: not measured (size-unknown): no figure for width_mm, depth_mm, above_grade_mm, section_mm; 0 of 25 pieces carry a known force
post@3048,2438,-1219: ground reaction unknown (not measured (size-unknown): no figure for width_mm, depth_mm, above_grade_mm, section_mm)
post@3048,4876,-1219: ground reaction unknown (not measured (size-unknown): no figure for width_mm, depth_mm, above_grade_mm, section_mm)
post@610,2438,-1219: ground reaction unknown (not measured (size-unknown): no figure for width_mm, depth_mm, above_grade_mm, section_mm)
post@610,4876,-1219: ground reaction unknown (not measured (size-unknown): no figure for width_mm, depth_mm, above_grade_mm, section_mm)

HEADER UTILISATIONS

status: known; largest 0.61
door@610,4819: utilisation 0.08; bending 0.61 MPa, shear 0.15 MPa, deflection 0.05 mm; load 4.44 N/mm against capacity 57.63 N/mm
glazing@610,2495,1219: utilisation 0.08; bending 1.29 MPa, shear 0.16 MPa, deflection 0.38 mm; load 2.33 N/mm against capacity 28.82 N/mm
glazing@610,2495,2438: utilisation 0.61; bending 9.7 MPa, shear 1.2 MPa, deflection 2.89 mm; load 17.6 N/mm against capacity 28.82 N/mm

BUILD SIMULATION TOTALS

ended quiescent; clock step 1 minute until 1440
finish_time = 22
labour_hours = 14
learning = 0.6
idle = 122
score = -119.4
final stocks:
budget_usd = 16069.77
cost_usd = 3930.23
floor_panel_ft2_set = 63.98
pieces_not_priced = 5
post_holes_dug = 4
roof_panel_ft2_set = 216.01
wall_panel_ft2_set = 366.02

PROJECT	SHEET	SHEET NUMBER	DATE
Small house on posts, from Jim's pencil sketch	**Design summary**	S-1 (page 1 of 2)	2026-10-20
CONFIG COMMIT	ENGINE PIN	SCALE	SOURCE
7dbc993	we 073b001	not to scale	world-engine geometry; no line is drawn by hand